

BENI-MELLAL, MoroccoWMO: **601910**Lat: **32.367N**Lon: **6.400W**Elev: **472**StdP: **95.78**Time Zone: **1.00 (E01)**Period: **94-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	2.5	-2.2	3.3	13.3	-0.7	3.8	10.8	6.6	11.2	5.6	11.3	1.1	120	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	41.3	20.3	39.4	20.4	37.4	20.4	22.8	34.9	22.1	34.1	21.5	33.4	3.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.8	27.4	18.5	14.1	26.7	17.8	13.5	26.1	69.6	34.9	66.9	34.2	64.7	33.6	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.2	4.3	-0.4	43.5	1.7	1.3	-1.7	44.4	-2.7	45.1	-3.7	45.9	-4.9	46.8		
			-1.7	24.1	2.0	1.0	-3.1	24.9	-4.2	25.5	-5.3	26.1	-6.7	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	10.7	12.6	15.2	17.5	20.6	25.3	28.7	28.2	24.4	20.7	15.1	11.9
	DBStd	6.75	2.34	2.62	2.99	2.80	3.71	3.05	3.16	2.59	2.96	2.75	3.23	2.33
	HDD10.0	37	19	7	2	0	0	0	0	0	0	0	1	8
	HDD18.3	884	237	162	105	47	18	0	0	0	1	9	107	199
	CDD10.0	3421	41	79	162	226	328	459	579	563	433	331	154	68
	CDD18.3	1224	0	0	7	23	88	209	321	304	183	81	9	0
	CDH23.3	15827	2	23	201	424	1103	2562	4496	4015	1978	864	157	1
CDH26.7	8639	0	1	39	146	532	1408	2819	2382	965	318	29	0	
Wind	WSAvg	2.0	1.4	1.6	1.7	2.0	2.2	2.4	2.5	2.5	2.2	1.8	1.5	1.5
Precipitation	PrecAvg	354	44	40	42	42	22	10	4	3	13	34	46	51
	PrecMax	626	156	115	162	152	84	67	21	12	74	117	181	142
	PrecMin	174	1	0	0	0	0	2	0	0	1	3	0	0
	PrecStd	110	40	30	38	39	25	15	6	4	17	27	48	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	26.0	30.1	33.5	39.0	42.1	43.9	42.6	39.7	34.3	29.7	22.6
		MCWB	12.7	14.1	15.9	17.2	18.6	20.1	20.3	20.2	19.9	17.8	16.0	13.5
	2%	DB	20.6	23.6	27.6	30.7	35.1	39.2	42.0	40.3	36.4	31.9	27.2	21.0
		MCWB	12.7	13.6	15.6	16.5	18.5	19.8	20.3	20.8	19.8	17.5	15.5	13.4
	5%	DB	19.2	22.1	25.6	28.0	32.5	36.6	40.0	38.2	34.2	30.1	25.2	19.6
		MCWB	12.2	13.6	15.0	16.1	18.3	19.9	20.6	20.8	19.6	17.3	15.1	13.2
	10%	DB	17.4	20.2	23.6	25.6	29.3	34.1	37.6	36.2	32.0	28.2	22.6	18.1
		MCWB	11.8	13.1	14.6	15.6	17.8	19.6	20.5	20.8	19.3	17.3	14.6	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.7	18.6	19.4	21.3	22.6	24.3	23.8	22.4	20.1	18.1	16.8
		MCDB	19.4	21.7	26.2	29.2	32.0	34.7	37.7	35.5	32.7	28.6	25.6	19.4
	2%	WB	14.0	15.5	17.2	18.0	19.8	21.7	22.9	22.8	21.5	19.3	16.8	15.3
		MCDB	18.3	20.4	24.6	27.3	31.0	33.3	35.5	34.8	31.6	27.3	23.9	18.1
	5%	WB	12.9	14.5	16.1	17.0	18.9	21.0	22.2	22.1	20.7	18.6	15.9	14.2
		MCDB	17.6	19.8	23.0	25.4	29.9	33.0	34.7	34.2	30.7	26.2	22.8	17.8
	10%	WB	12.0	13.5	15.2	16.1	18.0	20.3	21.5	21.5	20.1	18.0	15.0	13.3
		MCDB	16.5	19.0	21.6	24.1	28.0	31.5	33.9	33.4	29.5	25.3	21.4	17.2
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.2	13.7	13.4	13.9	15.3	16.2	15.3	13.1	13.5	13.6	12.5
		MCDBR	16.5	17.8	18.3	19.0	19.2	19.4	18.7	18.1	16.9	17.2	17.5	15.6
		MCWBR	9.3	9.1	8.0	7.2	6.0	5.2	4.6	4.8	5.0	6.1	7.5	8.5
	5% WB	MCDBR	13.8	14.3	15.7	16.2	16.6	16.8	16.3	15.9	14.0	13.3	14.5	11.6
		MCWBR	8.4	8.1	8.0	6.8	5.8	5.3	4.9	4.9	4.6	5.2	6.6	7.0
Clear-Sky Solar Irradiance	taub	0.291	0.335	0.394	0.410	0.451	0.495	0.562	0.538	0.453	0.408	0.339	0.303	
	taud	2.541	2.405	2.216	2.201	2.094	1.977	1.806	1.879	2.116	2.242	2.456	2.535	
	Ebn at Noon	940	920	882	880	844	803	748	759	813	825	870	905	
	Edh at Noon	83	105	137	145	162	182	215	197	149	122	90	79	
All-Sky Solar Radiation	RadAvg	3.45	4.31	5.46	6.42	7.10	7.82	7.64	6.93	5.86	4.58	3.56	3.16	
	RadStd	0.30	0.40	0.34	0.40	0.45	0.35	0.21	0.25	0.29	0.23	0.26	0.24	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		+0.24	N/A	-1.23	+0.87	-0.54	-0.46	N/A	N/A	+85

Nomenclature: See separate page